Sainava Modak

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Education

Amity University, Noida **2023 – 2027**

*B.Tech, Computer Science and Engineering; CGPA: **9.51**; 100% Merit Scholarship*

Technical Skills

Programming: Java, Python, C++, JavaScript
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems
Backend & Databases: Spring Boot, Node.js, Kafka, WebSockets, PostgreSQL, MySQL, MongoDB, Redis
Artificial Intelligence & ML: PyTorch, Transformers, YOLO, Graph Neural Networks (GNN), Bi-LSTM, RAG, OpenCV
Cloud & Tools: AWS (Cloud Practitioner), Docker, CI/CD, Git, Postman

Projects

LingoStream | Java, Spring Boot, Redis, PostgreSQL, WebSockets **GitHub**

- Built highly concurrent distributed backend using Spring Boot and Redisson distributed locks for state synchronization, implementing a Redis cache-aside pattern to prevent data collisions and reduce read latency to under 5ms.
- Engineered real-time collaborative editing using STOMP WebSockets for full-duplex broadcasting, securing the Docker-containerized API architecture with stateless JWT authentication.

CV-Graph-RAG: Structural Literature Synthesis | Python, Neo4j, Llama 3, Flask **GitHub**

- Engineered a hybrid GraphRAG pipeline fusing a Neo4j knowledge graph with vector search, eliminating cross-document hallucinations and establishing deterministic multi-hop relational integrity across literature.
- Built an end-to-end processing pipeline featuring structured entity resolution, localized inference via Ollama, and an interactive dashboard displaying explicit, citation-backed evidence paths.

Tactical GNN Spatial Engine | Python, PyTorch, React, Graph Neural Networks **GitHub**

- Engineered a spatial sports analytics engine using PyTorch and Graph Attention Networks (GATv2) to process raw tracking data and model dynamic spatial dominance.
- Built an interactive React dashboard utilizing Voronoi polygons and custom kinematic vector visualizations to map real-time player momentum, acceleration, and pitch control.

Experience

GeoGlyph Inc. Japan **Remote**

AI Engineer Intern *June 2026 – Present*

- Building a real-time Edge AI prototype for the MOTiON platform using PyTorch, YOLO, and OpenCV, focusing on inference pipeline optimization to minimize latency during complex spatial modeling.

Open Health Systems Laboratory (OHSL) **Remote**

Software Engineering Intern *May 2026 – June 2026*

- Engineered a distributed edge gateway for the TELESYNERGY platform, implementing concurrent WebSocket pipelines and FHIR data normalization to guarantee synchronized, high-throughput data transmission across complex networks.

Awards & Open Source

- **Talent for Japan 2025 Winner (Top 11 of 5,000+ applicants):** Represented India in Japan; presented AI-driven Smart Factory Intelligence solutions to Japanese corporate stakeholders and media. **Media Coverage**
- **Open Source Contributions:** Engineered Sapiens2 masked MSE loss in Hugging Face Transformers to enable model fine-tuning; contributed to GirlScript Summer of Code and Hacktoberfest. **GitHub**

Research

Multimodal Deepfake Detection (Ongoing) | Python, PyTorch, OpenCV, Bi-LSTM **GitHub**

- Built multimodal deepfake detection pipeline using RGB, optical flow, and Bi-LSTM.
- Achieved 97% accuracy on FaceForensics++ and 95%+ on CelebDF datasets.

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- **AWS Certified Cloud Practitioner**
- **Python for Data Science (NPTEL)** — Top 5%